Welcome to Building Generator (v1.5) for Falcon BMS !

The **Building Generator** is a powerful tool designed for Falcon BMS developers and contributors. It allows you to seamlessly transform geographical data from OpenStreetMap, QGIS, or other GIS systems, into Falcon BMS-compatible files. These files can then be loaded into the Falcon BMS simulator, enriching the virtual environment with realistic structures.

Key Features:

1. **GeoJSON Support**:
    - Load building structures in GeoJSON format.
    - Seamlessly import complex building geometries.
2. **Falcon BMS Integration**:
    - Generate Falcon BMS files (Editor format) containing building information.
    - Inject Buildings directly into BMS using Sophisticated algorithm
    - Enhance your virtual world by adding accurate, detailed structures.
3. **Files and Folders Manager**:
    - The software efficiently manages its own files and folders.
    - Simplify your workflow with an organized structure.
4. **Customization**:
    - Filter, sort, and select buildings based on general characteristics.
    - Configure preferences through the user interface for seamless data injection.

Whether you’re creating realistic urban landscapes, military bases, or entire cities, the Building Generator streamlines the process, making it a valuable asset for both hobbyists and professional developers.

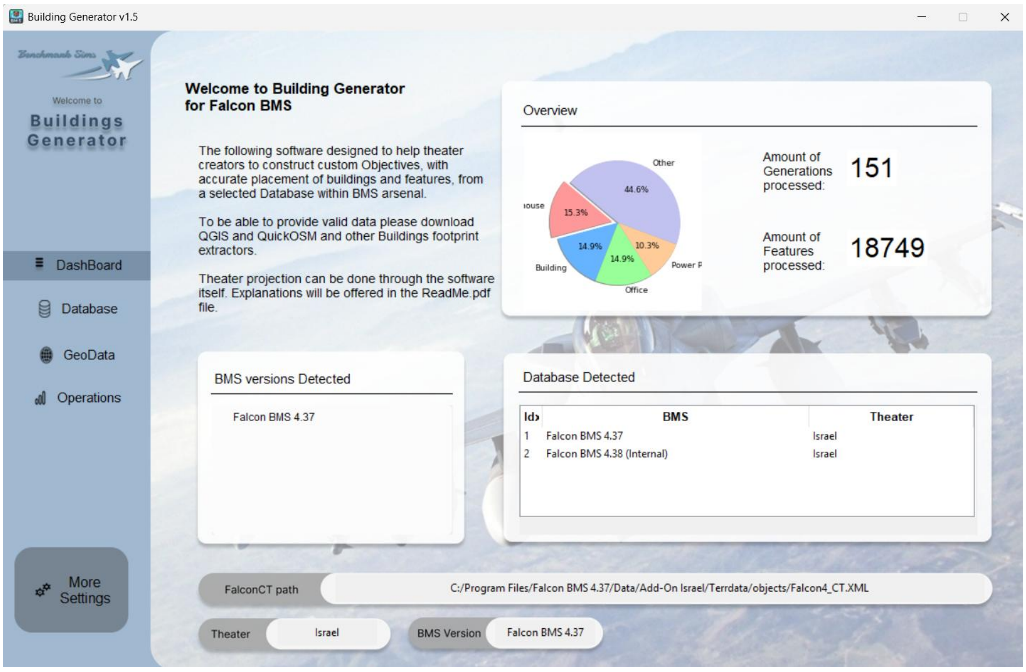
Let's Start

Installation is made by extracting the files from the compressed Zip file. after that, launching the Exe file of the software will open the Graphical user interface that you will work with.

Requirments

- Add exception for Antivirus (Some are sensitive for Exe made by python)
- Basic knowledge of OpenStreetMap keys and features
- Knowledge in OverPass-Turbo, Qgis with QuickOSM or Buildings footprint Extractor
- any kind of Text softwares (Notepad++ is recommended)
- Knowledge of Mission Commander, Falcon BMS Editor, Objectives viewer and how the Database look like

DashBoard - First Screen



After opening the software, you'll notice three main areas:

1. **Left Toolbar**:
   - Allows you to navigate between four different screens and open the settings window.
2. **Lower Part of the Screen**:
   - Provides information about the selected Class Table (CT), along with relevant details about the theater and BMS version that have been detected.
   - Custom information may also be displayed based on the current page.
3. **Main Screen**:
   - Displays relevant data for each page.

Overview

The overview section presents graphic and statistical information about the data that has been gathered since the beginning of usage.

BMS Versions Detected

In this window, you can view all the detected simulator versions of BMS that are recognized by Windows. The information is retrieved directly from the registry.

Database Detected

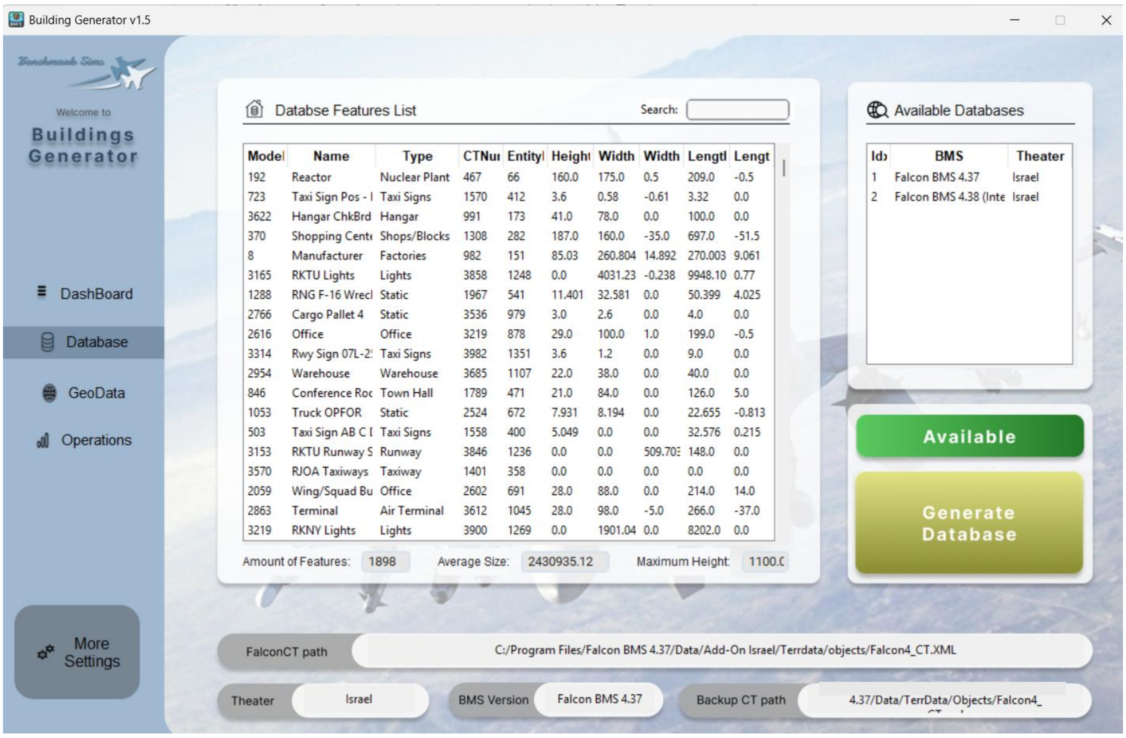
This table lists all the databases that have been generated by the software.

FalconCT Path

At the lower part of the screen, you can select your “base of operations,” which is the Class Table XML file. This file enables the software to collect essential information such as paths, theater details, BMS version, and available databases. The path will be displayed across all main screens, ensuring clarity about the data source.

> Note: The CT path is limited to Main Theatres (those containing CT/FCD/Parents information). Sub-theatres without proper databases are not supported.

Database - Second Screen



To enable the injection of structures into Falcon BMS, our software requires its own database. This database serves as the foundation for calculating, measuring, and injecting structures and features from the selected theater into Falcon BMS.

to be able to do that, locate the Class Table XML file for your desired theater. For example, the Korea Theater (KTO) Class Table XML file is located at:

> "Falcon BMS 4.XX\Data\TerrData\Objects\Falcon4_CT.xml"

- If a database is available for the theater, it will be auto loaded, indicated by a green image labeled "Available." Otherwise, a red image will indicate "Not Available."

Backup Class Table

Starting from Falcon BMS 4.37U4, a new feature allows drawing models and textures directly from the KTO default folder. This approach helps theaters save storage space, allowing dedicated storage for unique features.

If a certain theater is using this data redirection, a backup Class Table is needed to fetch the original models from the Korea theater. Clicking on "**Backup CT Path**" bar will open a window to select the Class Table of KTO (unless the code evolves into a more complex structure, allowing any theater to serve as its base).

Generating Database

After selecting the desired Class Tables, clicking on the yellow button labelled **"Generate Database"** will produce the calculated information and show it on the list of buildings.

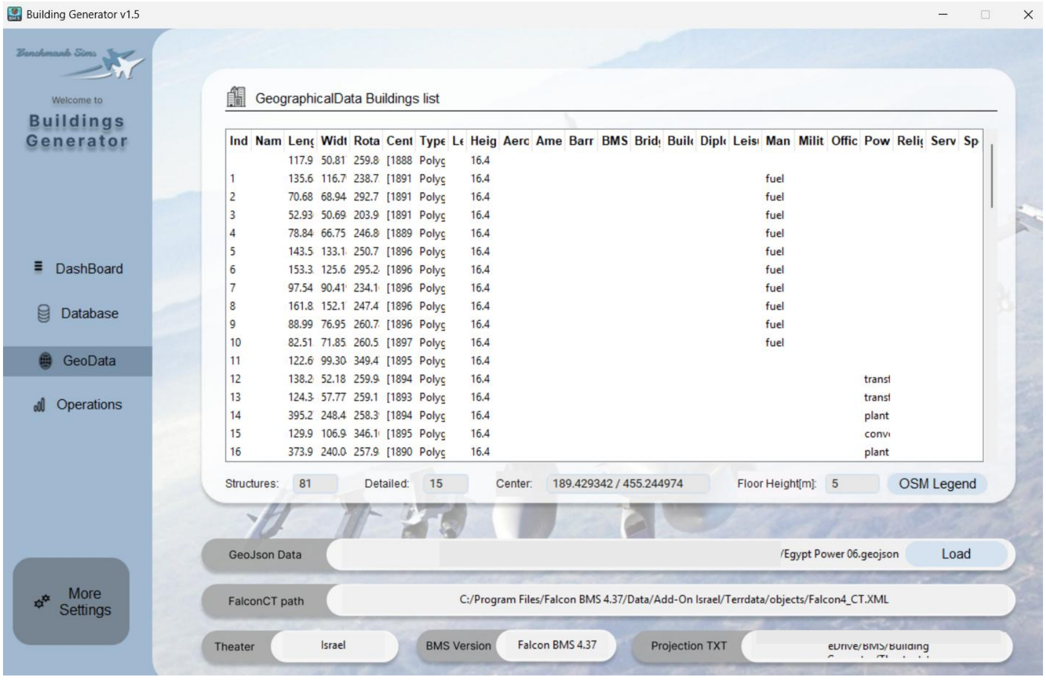
List of buildings

The information from the calculated database will be presented here. sorting the columns is possible by clicking on the desired title. And searching by name or number can be done by the search bar above it.

The titles are as follows:

> "ModelNumber" - Parent Number of the "Normal" variant
> "Type" - Classification of the model (Warehouse, Bunker, Ammo Dump etc)
> "CTNumber" - The Associate Class Table number
> "EntityIdx" - The Associate Feature Class Table number
> “Height” – The height in feet of each structure
> "Width" - Width of the fitted bonding box (shorter face) in feet
> "WidthOff"- Offset from the center on the same axis of
> "Width" in feet "Length" - Length of the fitted bonding box (longer face) in feet
> "LengthOff" - Offset from the center on the same axis of
> "Length" in feet "Height" - height in in feet
> "LengthIdx" - Index of longer face

GeoData- Third Screen



This section is dedicated to loading and viewing geographical data in **GeoJSON** format, which can be generated by GIS software (such as QGIS) or Overpass-Turbo.

To load this information, click on the "**GeoJSON Data**" bar. A selection window will allow you to pick the GeoJSON file containing the geographical data. It's essential to understand that Falcon BMS uses projection formulas to span it's world. If we haven't applied any projection calculations to our source data, we **must** force it through the software. By pressing the **Load** button, the information will be processed and displayed in the list of features.

Projection

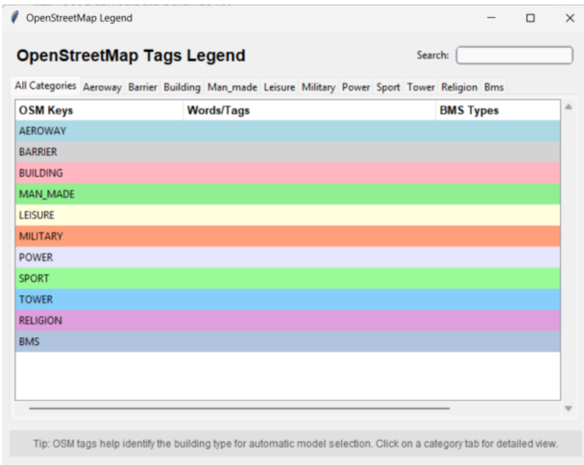
To implement a projection in the software, we need to first understand the boundaries of our theater. Starting from Falcon BMS 4.38, the projection equations are more accurate and aligned with modern standards.

Example of Korea Projection:

> Projection string=+proj=tmerc +lon_0=127.5 +ellps=WGS84 +k=0.9996 +units=m +x_0=512000 +y_0=-3.74929e+06

To load the projection, click on the **"Projection TXT"** bar. This will open a window where you can select a text file containing a custom string like the one shown above.

Open Street Map Legend



The software is equipped with a built-in implementation of the OpenStreetMap (OSM) key map, tailored to parallel BMS structures. Pressing the **"OSM Legend"** button opens a window displaying all key items and their translations to words and types. Click the tabs or use the search bar to find the supported titles.

- Words are searched through the "Names" of the structures.
- Types are searched based on their definitions in the database.

List of features

The information from the loaded geographic file will be presented here. You can sort the columns by clicking on the desired title. The tittles are as follows:

> "Index", "Name", "Length", "Width", "Rotation", "Center", "Type", "Levels", "Height", "Aeroway", "Amenity", "Barrier", "Bridge", "Building", "Diplomatic", "Leisure", "Man Made", "Military", "Office", "Power", "Religion", "Service", "Sport"

Below the table, you'll find details such as the total number of features, the calculated center of all features, and the count of "Detailed" features. These detailed features have more accurate descriptions.

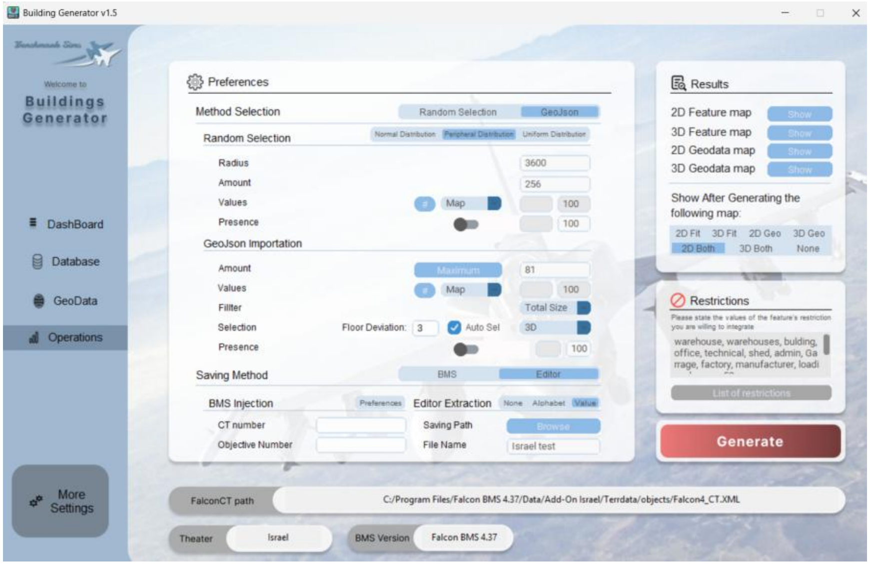
For example:

> Military-> Bunker
>
> Aeroway -> Terminal

Floor Height

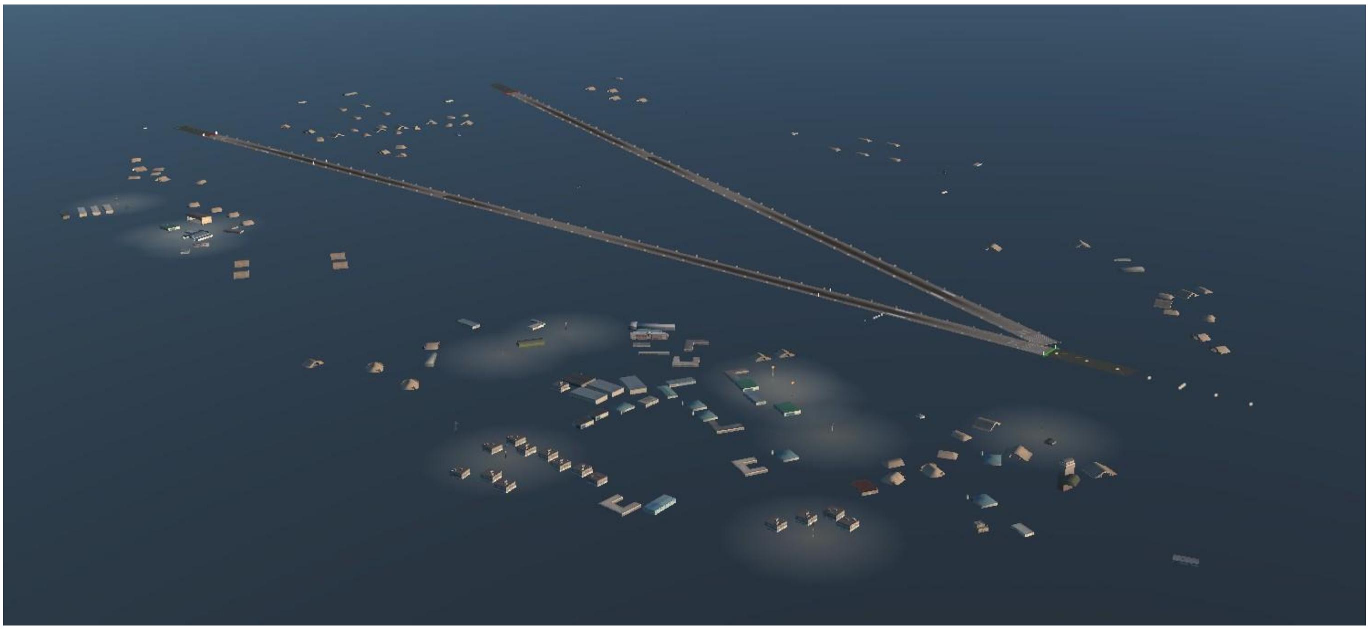
The user now can determine the floor height that would be calculated in the program. Buildings without height or floor count would considered having only one floor, which will determine their final height for the fitting algorithm later on.

Operations- Last Screen



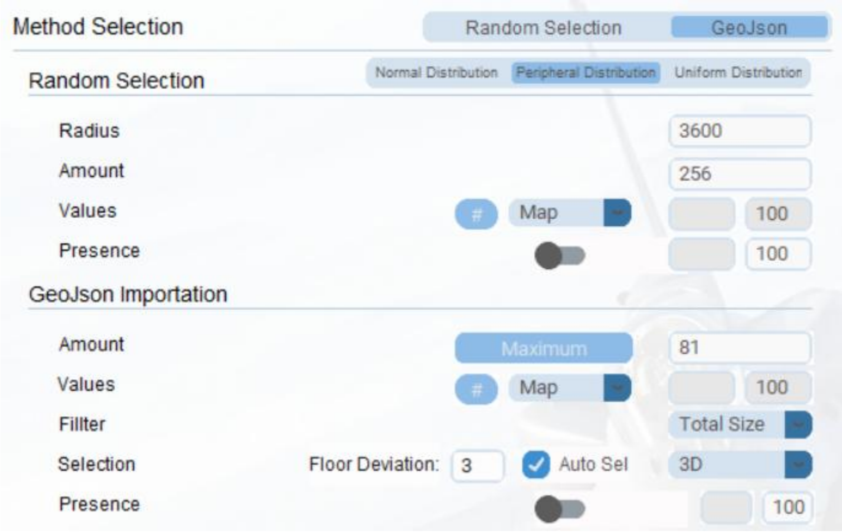
In this final section, we'll discuss how to produce compatible BMS files that can be directly loaded into the BMS editor. These files allow you to assign new objectives with different configurations of buildings and features.

Example of an airbase which generated (mostly) by Building Generator:



Preferences

This area is divided into two main sections: **Method of Selection** and **Saving Method**.



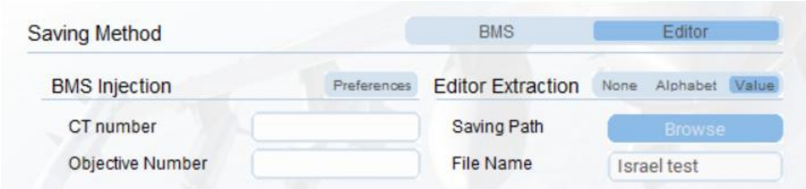
- **Method of Selection**:
    - Random Selection - will place structures randomly in a desired Radius. Amount, values and Presence are needed to complete the demands of the simulator.

      Sub Algorithms for randomizations are available:

      - Normal Distribution
      - Peripheral Distribution
      - Uniform Distribution

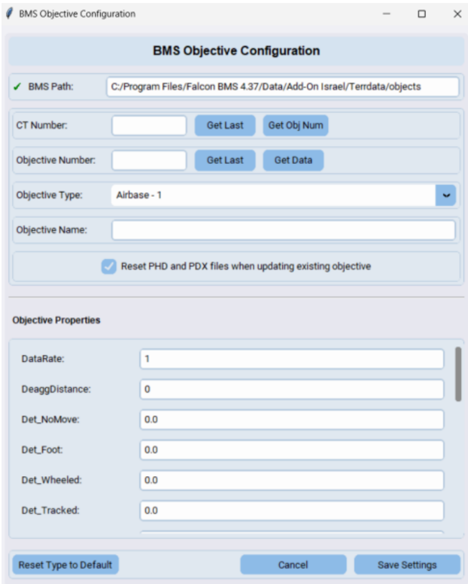
    - GeoJson Importation - Will take the Geodata that been imported in GeoData page, filter and select relevant structures through the criteria provided in OSM and the Restrictions window.

      - **Filter** - will filter the list of features according to the number of features written in the "Amount" entry. The filter will not be applied if the number of features is bigger then the actual number of features loaded into the Geodata page.
      - **Selection**- fit for each geographical feature a suited structure from BMS. 3D and 2D are different by the factor of height. Auto checkbox will specificly reduce the amout of options to be more perciece to the detailed geographical feature based on the OSM legend.
      - **Presence** - Determines the chance of the feature appearing in the BMS world (0 - 100%).
      - The switch next to the values allows randomization within the specified integer range.
      - **Value** - Represents the importance level of a structure within an objective (0-100). Three options are available: “Solid,” “Random” (like “Presence”), and **Map** (to be discussed later).
      - **Floor Deviation** – Will deviate additional floor count to Structures based on normal distribution. For example, Floor Deviation == 3 will grand the possibility to increase in certain probability the structures by maximum of 3 floors.



- **Saving Method**:
  - Editor Extraction - Exports an Editor file that can be loaded into the Objective viewer, displaying the 3D map of new structures. *Note: Each objective is limited to 256 features*. Specify the saving path and file name to complete the saving process
  - BMS Injection –Exports the generated features directly into BMS Database. To do so, we need to provide the right information to the BMS Objective configuration window.

- **BMS Objective configuration**



To inject data into BMS, open the BMS Objective configuration by clicking on Preferences button. By doing so, the program will read and calculate templates for each objective criterion. Then the following data is needed:
-CT Number
-Objective Number
-Objective Type
-Objective name

After providing that information, we can save and using the injection method.

The window will offer more knowledge such as, the ability to get the last CT or Objective Number that is not been used, grabbing the Class Table Data from the Typed Objective Number, re-calculate the Median values from the Database of selected objective Type and so on.

Values Mapping

Pressing the **(#)** button in the "Value" row opens a map of values associated with each type. These values will be automatically assigned to a feature when injected into BMS. Save these values as a JSON file in the software folder by clicking **"Save"**. Saving will override existing values.

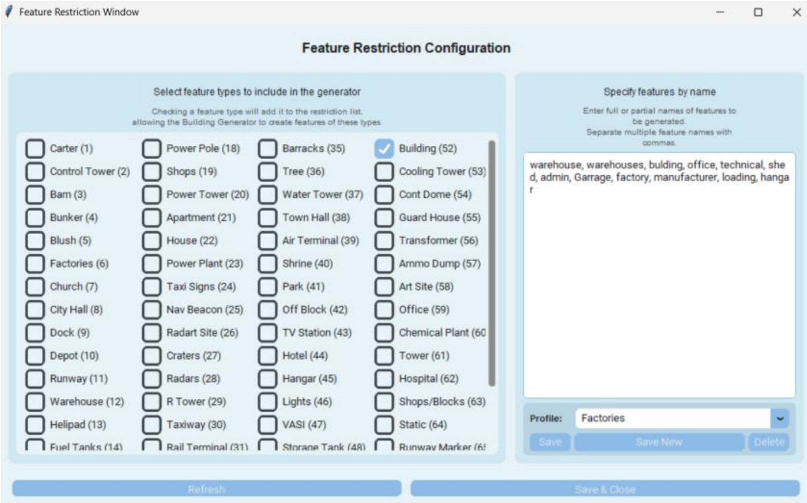
Use **Set Default** to return values to their default settings. Remember to save before exiting the window.

Restrictions

The purpose of this area is to limit the database from which feature selection is made

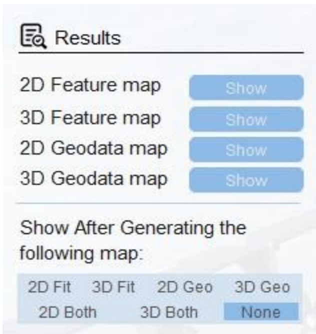
> For example, when importing a city, you may need specific types of buildings (e.g. shops, apartments, city halls) but not others (e.g., bunkers, cooling towers, ammo dumps). Each objective can be customized by restricting the types and names of structures needed.

The selection process involves specifying the full or partial names of the features we need or typing accurate building types. To assist with typing the types, the Restriction window is created:



- Pressing the "**List of restrictions**" button opens a window displaying all the types in Falcon BMS.
- Each entry or clicking on **"Refresh"** Button will load the corresponding types into the checkboxes,
- Exporting via the "**Save and Close**" button transfers the selected types back to the textbox in the main user interface. Full words are not affected.
  - Profiling is now available in the restriction window; it would save the textbox and checkboxes for later use. This way, customizing structures for certain types of objectives would be much easier and flexible.

Results



The Results area allows you to view 3D or 2D graphs of the structures that have been filtered and selected by the software. An auto showing option is available at the bottom.

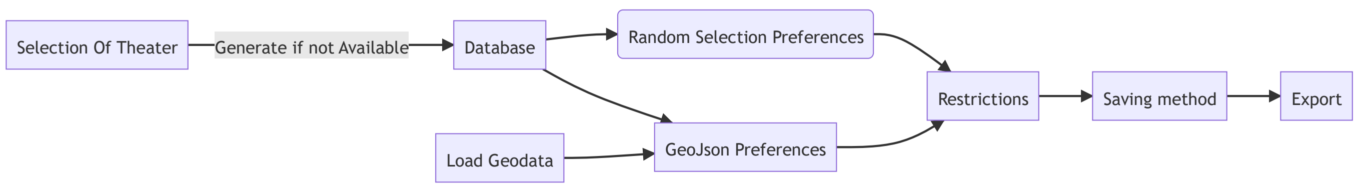
Generating the final file for Editor Extraction

Pressing the red button labelled "**Generate**" will calculate all the previous data, filter and fit structures, and finally save the features map to BMS or as an Editor file.

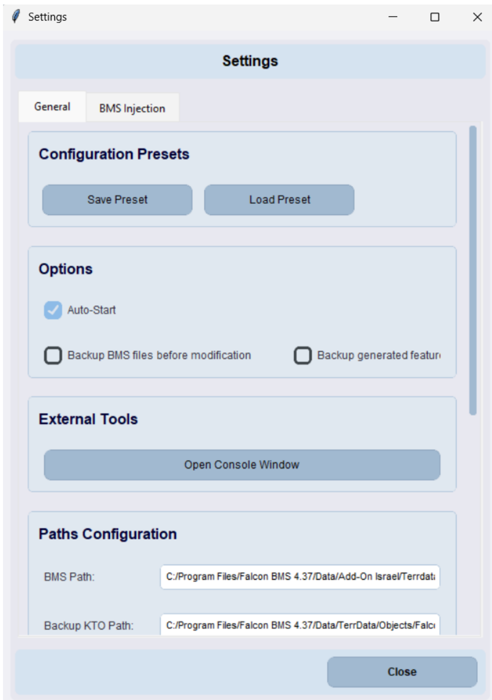
Example:

```
# BMS-BuildingGenerator (v0.95b) for FalconEditor - Objective Data

# Objective original location in Falcon World (Falcon BMS 4.38 with New Terrain)
# ObjX: 192.43972366007833
# ObjY: 360.77481036183394

Version=6

# FeatureEntries 172

FeatureEntry=2910 657.7566 -1700.3263 0.0000 266.3086 0000 0000 -1 100# 0) Watchtower
FeatureEntry=3861 -1990.0992 -1296.7921 0.0000 277.8533 0000 0000 -1 100# 1) Technical Shed
FeatureEntry=2910 6944.2007 -115.6840 0.0000 227.8624 0000 0000 -1 100# 2) Watchtower
FeatureEntry=2910 -3114.6553 -2481.5044 0.0000 353.9910 0000 0000 -1 100# 3) Watchtower
FeatureEntry=2910 -4915.7263 -2306.3714 0.0000 353.1572 0000 0000 -1 100# 4) Watchtower
FeatureEntry=3861 -4122.1246 -789.5436 0.0000 279.0903 0000 0000 -1 100# 5) Technical Shed
FeatureEntry=2910 2854.5418 -1221.0457 0.0000 355.2364 0000 0000 -1 100# 6) Watchtower
FeatureEntry=3861 3465.2067 -582.2963 0.0000 273.8141 0000 0000 -1 100# 7) Technical Shed
FeatureEntry=2910 5352.1722 -1903.7572 0.0000 192.5288 0000 0000 -1 100# 8) Watchtower
FeatureEntry=2910 3976.2268 -1105.2099 0.0000 213.1785 0000 0000 -1 100# 9) Watchtower
FeatureEntry=2706 -5086.8934 -1545.8474 0.0000 181.2189 0000 0000 -1 100# 10) Public Work Shed
FeatureEntry=3861 -3205.0342 -1614.3558 0.0000 342.8973 0000 0000 -1 100# 11) Technical Shed
FeatureEntry=2706 -4407.7187 -1780.9108 0.0000 217.4054 0000 0000 -1 100# 12) Public Work Shed
FeatureEntry=3861 -2016.5174 -953.6470 0.0000 273.3665 0000 0000 -1 100# 13) Technical Shed
FeatureEntry=3861 3491.6755 -927.6622 0.0000 273.2705 0000 0000 -1 100# 14) Technical Shed
FeatureEntry=3861 4617.1096 1592.3095 0.0000 266.7295 0000 0000 -1 100# 15) Technical Shed
FeatureEntry=3861 2647.1572 150.9167 0.0000 228.6914 0000 0000 -1 100# 16) Technical Shed
FeatureEntry=3861 4385.4837 1598.8473 0.0000 351.6743 0000 0000 -1 100# 17) Technical Shed
FeatureEntry=2586 4669.1903 1824.8292 0.0000 354.2894 0040 0000 -1 100# 18) Ammo Dump
FeatureEntry=2586 4579.8158 2049.5128 0.0000 352.4054 0040 0000 -1 100# 19) Ammo Dump
```

Flow of actions:



Settings



1. **Presets:**

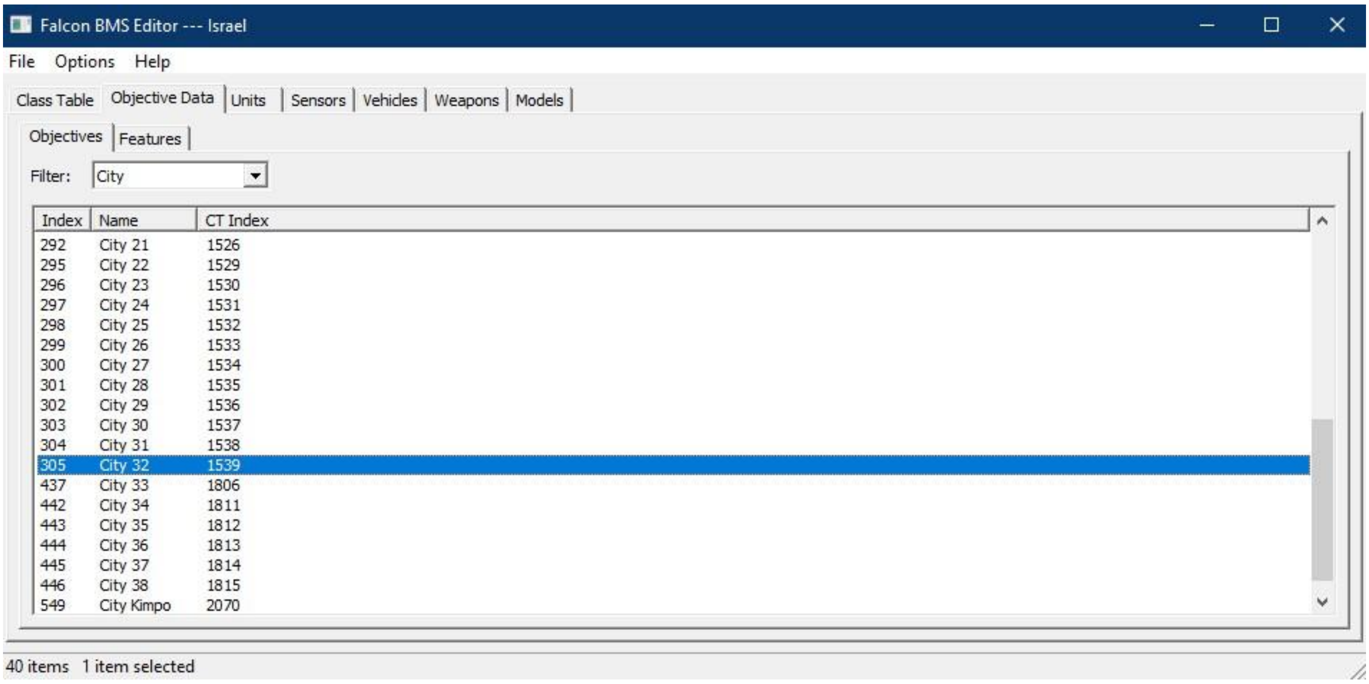
    - User interface presets are salvable and can be loaded on startup if the checkbox is selected. This allows you to customize your preferred layout and settings for a smoother experience.

2. **Options**:
    - AutoStart – Loads Preset on startup
    - Backup BMS files before Modification – Save the crucial files of BMS before injecting Features into BMS
    - Backup Generated Features – Save the final results in Generated folder for backup
3. **External Tools**
    - Console window – will redirect logs and prints to external console for inspection over the flow of code.
4. **Path Configuration**
    - Shows all important paths in the program.
5. **Logging Configuration**

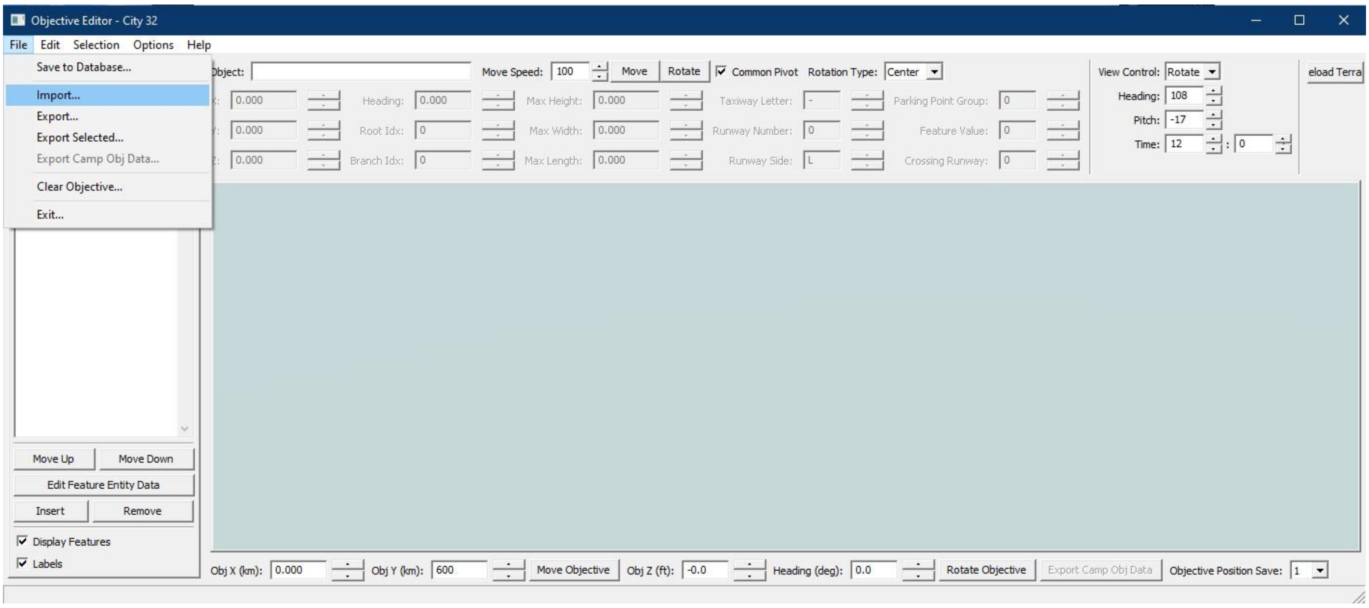
    - Enable selection of multiple logging options such as INFO, DEBUG, CRITICAL.
    - Providing the ability to decide where to show the logs, in logs folder or in the external console.
6. **Injection**
    - Injection tab contains all the data of Class Tables and Objectives that been calculated by the Objective Configuration window. It would allow to edit, recalculate and save them individually.

Loading Structures to BMS

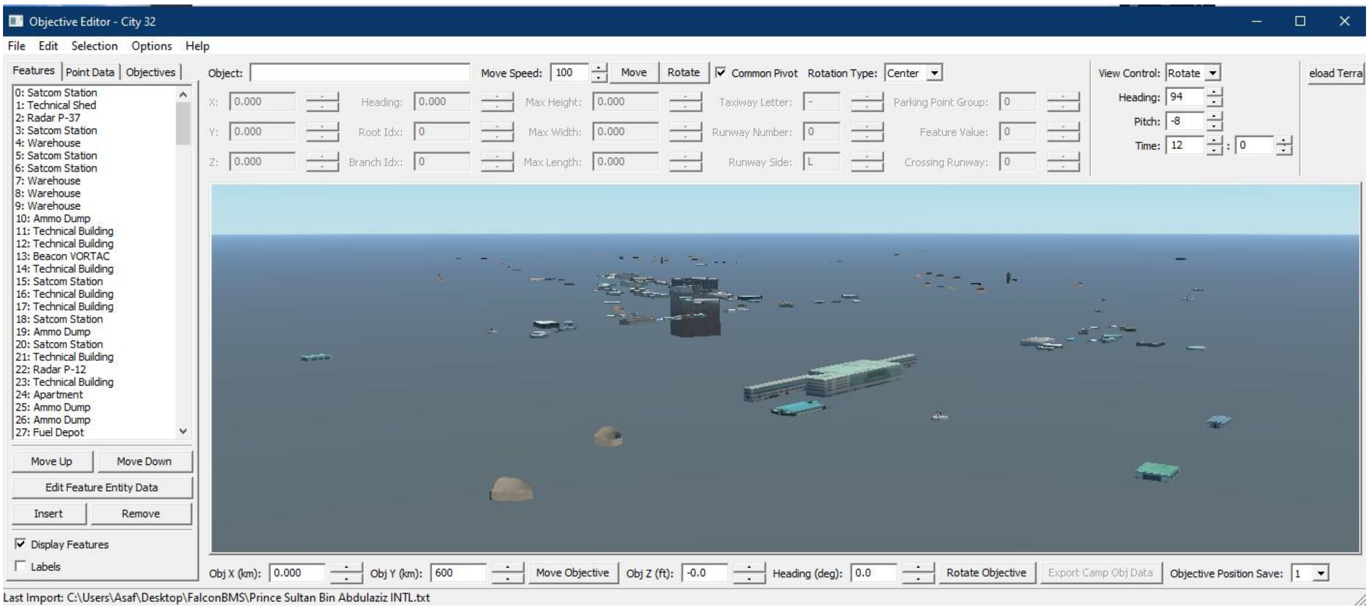
1. After generating the final file, launch the BMS editor from the BMS launcher.
2. Select or create an objective by double-clicking on it.



3. The Objective viewer/editor should open. Go to **File** -> **Import**.
4. Select the generated file and load it into the viewer.



5. Congratulations! You should now see the new map of features.



6. The next steps involve manually refining and correcting any issues you identify. Adjust the relevant PointData or other specifications of the objective before saving it and moving on to the next one.

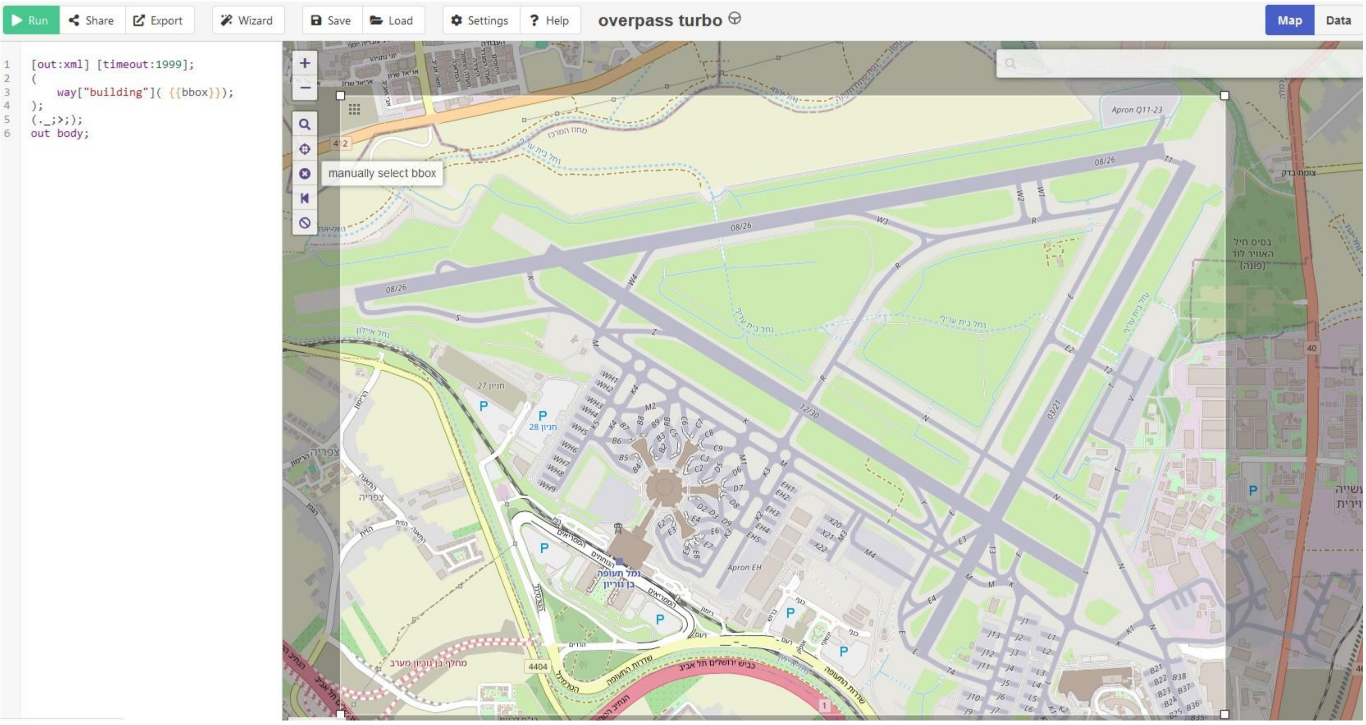
Demonstration of Basic Geographical Data

We will use Overpass-Turbo to demonstrate basic exportation of buildings from Ben-Gurion INTL Airfield.

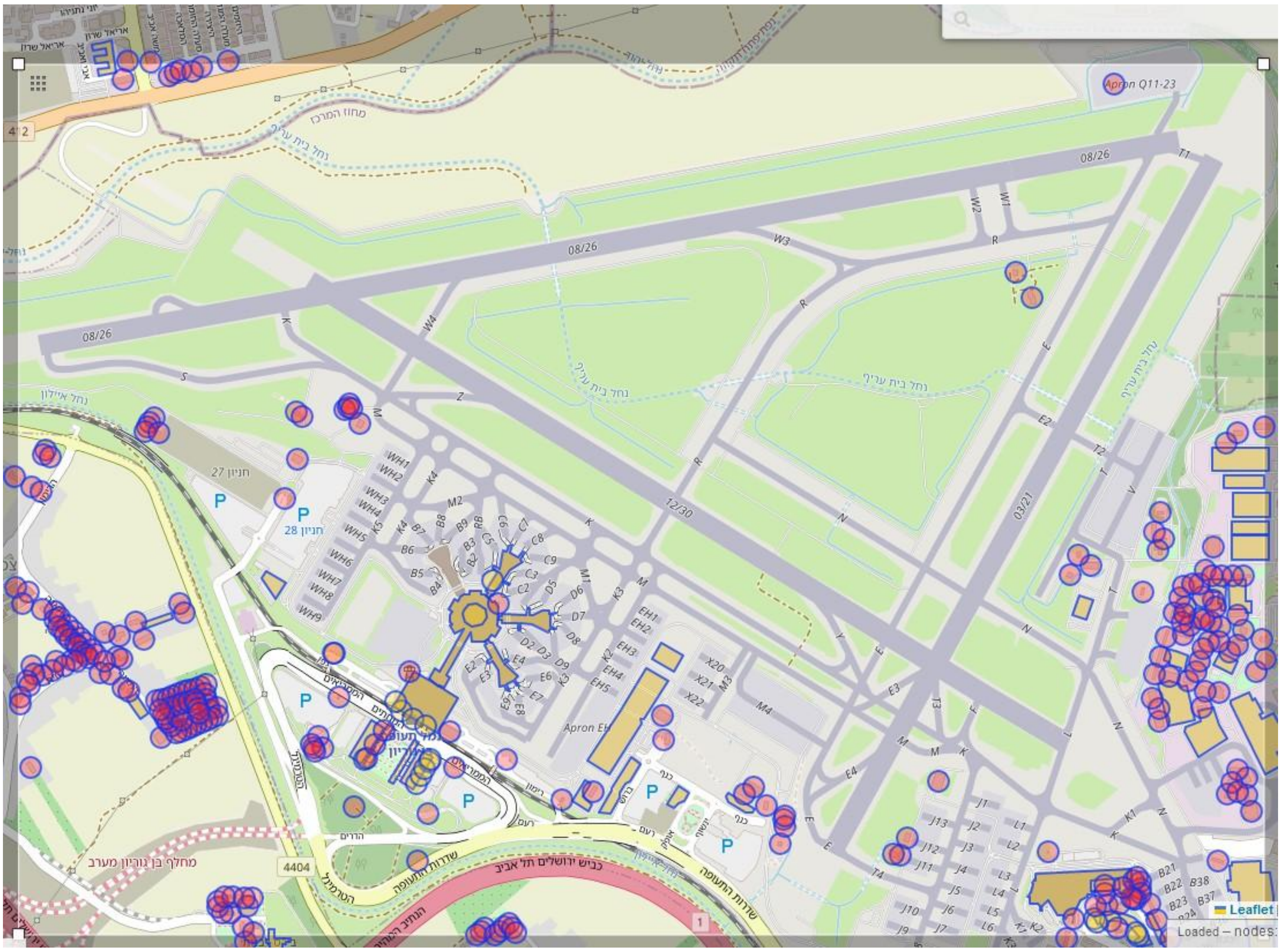
1. Go to the URL Overpass Turbo.
2. Find your center of interest (e.g., Ben-Gurion INTL Airfield).
3. Press the **selection box** and assign the region of interest.
4. Insert the following text into the left console:

```
[out:xml] [timeout:1999];
(
    way["building"]( {{bbox}});
);
(._;>;);
out body;
```

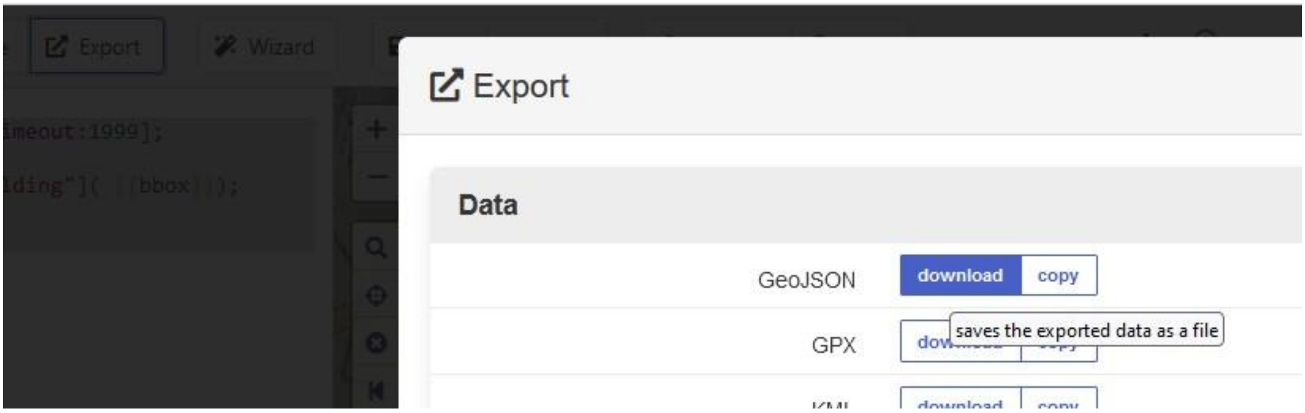
The screen should look like this:



6. Press "**Run**" in the top left area of the website. After the loading screen finishes, buildings should be displayed



7. Press “**Export**” in the top left area of the website and select “Download” for the GeoJSON file



8. **Load** the downloaded file into “Building Generator” and continue with the procedure to export the structures map for BMS.